(Advance Stats) - Week-2

i) $P[a \le X \le b] = \int_a^b f_n(n)\,dn$

ii) $P[n \le a] = \int_{-\infty}^{a} f_n(n)\,dn$

iii) $P[n \ge a] = \int_a^{\infty} f_n(n)\,dn$

$F_n(a) = P[n \le a]$

$P[n \ge a] = 1 - F_n(a)$

$P[a \le n \le b] = F_n(b) - F_n(a)$

$$f_n(n) = \begin{cases} 2n & 0 \le n \le 1 \\ 0 & \text{else} \end{cases}$$

$$E[X] = \int_{-\infty}^{\infty} n f_n(n)\,dn = \int_0^1 2n^2\,dn = \left[\frac{2n^3}{3}\right]_0^1 = \frac{2}{3}$$

$$Var[n] = E[X^2] - (E[X])^2$$

$$E[X]^2 = \int_{-\infty}^{\infty} n^2 f_n(n)\,dn = \int_0^1 2n^3\,dn = \left[\frac{n^4}{2}\right]_0^1 = \frac{1}{2}$$

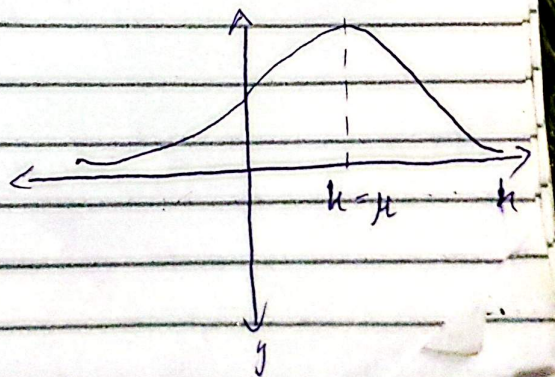
$$Var[n] = \frac{1}{2} - \frac{4}{9} = \frac{1}{18}$$

Normal or Guassian R.V

$$f_n(n) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2}\left(\frac{n-\mu}{\sigma}\right)^2\right\}$$

$$-\infty < n < \infty$$

$E[n] = \mu$, $Var[n] = \sigma^2$



$$F_x(a) = P(x \le a) = \int_{-\infty}^{a} \frac{x}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right\} dx$$

$$F_x(a) = P(X \le a) = P\left(\frac{x-\mu}{\sigma} \le \frac{a-\mu}{\sigma}\right)$$

$$F_x(a) = P\left(Z \le \frac{a-\mu}{\sigma}\right)$$

$$F_x(a) = F_Z\left(\frac{a-\mu}{\sigma}\right)$$

$X \sim N(\mu = 50, \sigma = 10) \rightarrow F_x(62) - F_x(45)$

$$P[45 \le X \le 62] = F_Z\left(\frac{b-\mu}{\sigma}\right) - F_Z\left(\frac{a-\mu}{\sigma}\right)$$

$$= F_Z\left(\frac{6}{5}\right) - F_Z\left(-\frac{1}{2}\right)$$

$$= F_Z(1.2) - F_Z(-0.5)$$

$$= 0.8849 - 0.3085$$

$$= 0.5764$$

$$f_x(x) = \begin{cases} 20000/x^3, & x > 100 \\ 0, & \text{otherwise} \end{cases}$$

$$E[X] = \int_{100}^{\infty} x \frac{20000}{x^{3/2}}\, dx = \left[\frac{20000}{1} x^{-1}\right]_{100}^{\infty} = \frac{20000}{100} = 200$$

$$P[X = 200] = 0$$

$$P[X > 200] = \int_{200}^{\infty} \frac{20000}{x^3} = -\frac{20000}{2}\left[x^{-2}\right]_{200}^{\infty} = 0.25$$

(AS - Week -3)

N. Sample:

$$Y = a_1 x_1, a_2 x_2, \ldots, a_n x_n$$

Pdf / Pmf of x_i are known:

$$\mu_i = E[x_i] \quad , \quad \sigma_i^2 = Var[x_i] \quad , \quad i = 1, 2, \ldots, N$$

$$\mu_i = E[a_1 x_1 + a_2 x_2 \ldots]$$

$$= E[a_1 x_1] + E[a_2 x_2] + 2\,[E[a_1 x_2]]$$
$$= a_1 E[x_1] + a_2 E[x_2] + \ldots + a_n E[x_n]$$
$$= a_1 \mu_1 + a_2 \mu_2 + \ldots + a_n \mu_n$$

$$\mu_y = \sum_{i=1}^{N} a_i \mu_i$$

$$\sigma_y^2 = Var[y]$$

→ If X_i are independant.

$$\sigma_y^2 = Var[y] = Var[a_1 X_1 + a_2 X_2 + \ldots + a_n X_n]$$
$$= Var[a_1 X_1] + Var[a_2 X_2] + \ldots + Var[a_n X_n]$$

$$= a_1^2 \sigma_1^2 + a_2^2 \sigma_2^2 + \ldots + a_n^2 \sigma_n^2$$

$$\sigma_y^2 = \sum_{i=1}^{N} a_i^2 \sigma_i^2$$

→ If X_i's are independant and identically distributed (IID)

$$\mu = \mu_i \quad , \quad \sigma^2 = \sigma_i^2 \quad , \quad i = 1, 2, \ldots, N$$

$$\mu_y = \mu \sum_{i=1}^{N} a_i$$
$$\sigma_y^2 = \sigma^2 \sum_{i=1}^{N} a_i^2$$

if, $a_i = 1$

$$\mu_y = N\mu$$
$$\sigma_y^2 = N\sigma^2$$

$X_1 \sim N(0,1)$, $X_2 \sim N(-1,2)$, X_1, X_2 ind

$y = x_1 + (-2)x_2 =$

$\mu_y = 0 + 2 = 2$

$\sigma_y^2 = (1)^2 + (-2)^2(2) = 9$

→ If X_i's are independent normal R.V, Y is also normal.

$Y \sim (2,9)$

$$f(y) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$$

Week 4 : A5

-> True mean of population = μ } unknown constraints
-> True variance of population = σ^2 }

N - samples : $X_1, \ldots, X_n$
all X_i's are IID(R.V) :

$$\mu = E[X_i] \quad , \quad \sigma^2 = var[X_i] \quad , \quad i = 1, 2, \ldots, N$$

Sample Mean ($\bar{X}$) is an estimation of μ

$$\bar{X} = \frac{1}{N} \sum_{i=1}^{N} X_i$$

$$\mu_{\bar{X}} = E[\bar{X}] = E\left[\frac{1}{N} \sum_{i=1}^{N} X_i\right] = \frac{1}{N} \sum_{i=1}^{N} E[X_i] = \frac{1}{N} \times N\mu = \mu$$

$$\boxed{\mu_{\bar{X}} = \mu}$$

-> Unbiased estimator of μ

$$\sigma_{\bar{X}}^2 = var[\bar{X}] = var\left[\frac{1}{N} \sum_{i=1}^{N} X_i\right] = \frac{1}{N^2} \sum_{i=1}^{N} var[X_i] = \frac{1}{N^2} \sum_{i=1}^{N} \sigma^2$$

$$= \frac{1}{N} \sigma^2 = \frac{\sigma^2}{N}$$

$$\boxed{\sigma_{\bar{X}}^2 = \frac{\sigma^2}{N}}$$

-> Variance reduces with increasing samples N.

Prob. Distribution of $\bar{X}$:

Using CLT :

$$P[\bar{X} \leq a] = P\left[Z \leq \frac{a - \mu_{\bar{X}}}{\sigma_{\bar{X}}}\right] = P\left[Z < \frac{a - \mu}{\sigma/\sqrt{N}}\right]$$

$$F_{\bar{X}}(a) = F_Z\left(\frac{a - \mu}{\sigma/\sqrt{N}}\right)$$

Ex 8.4) $\mu = 800$, $\sigma = 40$, $N = 16$,

$$P[\bar{X} \leq 775] = P\left[Z \leq \frac{775 - 800}{40/4}\right] = P[Z \leq -2.5] = 0.0062$$

Sample variance (S^2) is an estimator of σ^2

$$S^2 = \frac{1}{N-1} \sum_{i=1}^{N} (X_i - \bar{X})^2$$

$$E[S^2] = E\left[\frac{1}{N-1} \sum_{i=1}^{N} (X_i - \bar{X})^2\right] = \sigma^2$$

CLT ?

When all X_i's are IID Normal

20-1

$$\chi^2 = \frac{(N-1)S^2}{\sigma^2}$$ ← t-table ?

↓

Chi-Squared R.V with $\nu = N-1$ degrees of freedom

Week-5 - A5)

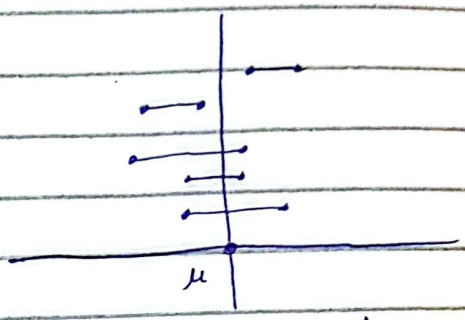
$A = [1, 10]$, $x = 5$

$P[x \in A] = 1$

$P[x \in A] = 0$, $x = 12$

Given that true variance is known, μ is unknown, confidence interval $100(1-\alpha)$

$$\left[\bar{X} - Z_{\alpha/2} \frac{\sigma}{\sqrt{N}} , \bar{X} + Z_{\alpha/2} \frac{\sigma}{\sqrt{N}} \right]$$



95% cross the unknown line
5% do not this means 95% C.I

$N = 36$

$\sigma = 0.3$

$\bar{x} = 2.6$

$\alpha = 0.05$ $1 - \alpha/2 = 0.95$

1.96

$= 2.6 - \frac{0.3}{\sqrt{36}}$

$\mu = (2.574 , 2.626)$ $(2.502, 2.698)$

$$\left(\bar{x} - Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} , \bar{x} + Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right)$$

$$\boxed{n = \frac{Z_{\alpha/2}\, \sigma}{e}}$$

If population normal we can find unknown variance

→ When population variance is unknown: C.I for Pop mean

$$\bar{x} - t_{\alpha/2} \frac{s}{\sqrt{n}} < \mu < \bar{x} + t_{\alpha/2} \frac{s}{\sqrt{n}}$$

$v = n - 1$ (Degrees of freedom)

→ Directly gives value of α no need for $1 - \alpha/2$

$s = 0.2828$ $n = 7$, $v = 6$

$\bar{X} = 20$

$\alpha = 0.05$, $\alpha/2 = 0.025$

$t_{\alpha/2} = 2.447$

$(9.74, 10.26)$

C.I for σ^2:

$$\frac{(n-1)s^2}{\chi^2_{\alpha/2}} < \sigma^2 < \frac{(n-1)s^2}{\chi^2_{1-\alpha/2}}$$

~~$s = 0.5349$~~, $s^2 = 0.2862$

$\bar{X} = 46.12$

$n = 10$, $v = 9$

$\alpha'/2 = 0.0.25$, $1-\alpha/2 = 0.975$

$\chi^2_{\alpha/2} = 19.023$

$\chi^2_{0.975} = 2.7$

$(0.1354, 0.954)$

FSE - Class